

3.7.1 Operating Characteristic Curves

Recall that an **operating characteristic (OC) curve** is a plot of the type II error probability of a statistical test for a particular sample size versus a parameter that reflects the extent to which the null hypothesis is false. These curves can be used to guide the experimenter in selecting the number of replicates so that the design will be sensitive to important potential differences in the treatments.

We consider the probability of type II error of the fixed effects model for the case of equal sample sizes per treatment, say

$$\begin{aligned}\beta &= 1 - P\{\text{Reject } H_0 | H_0 \text{ is false}\} \\ &= 1 - P\{F_0 > F_{\alpha, a-1, N-a} | H_0 \text{ is false}\}\end{aligned}\tag{3.43}$$

To evaluate the probability statement in Equation 3.43, we need to know the distribution of the test statistic F_0 if the null hypothesis is false. It can be shown that, if H_0 is false, the statistic $F_0 = MS_{\text{Treatments}}/MS_E$ is distributed as a **noncentral F** random variable with $a - 1$ and $N - a$ degrees of freedom and the noncentrality parameter δ . If $\delta = 0$, the noncentral F distribution becomes the usual (central) F distribution.

Operating characteristic curves given in Chart V of the Appendix are used to evaluate the probability statement in Equation 3.43. These curves plot the probability of type II error (β) against a parameter Φ , where

$$\Phi^2 = \frac{n \sum_{i=1}^a \tau_i^2}{a\sigma^2}\tag{3.44}$$

The quantity Φ^2 is related to the noncentrality parameter δ . Curves are available for $\alpha = 0.05$ and $\alpha = 0.01$ and a range of degrees of freedom for numerator and denominator.

Choice of Sample Size. Choosing the **sample size**, or the **number of blocks** to run, is an important decision when using an RCBD. Increasing the number of blocks increases the number of replicates and the number of error degrees of freedom, making design more sensitive. Any of the techniques discussed in Section 3.7 for selecting the number of replicates to run in a completely randomized single-factor experiment may be applied directly to the RCBD. For the case of a fixed factor, the operating characteristic curves in Appendix Chart V may be used with

$$\Phi^2 = \frac{b \sum_{i=1}^a \tau_i^2}{a\sigma^2} \quad (4.19)$$

where there are $a - 1$ numerator degrees of freedom and $(a - 1)(b - 1)$ denominator degrees of freedom.

EXAMPLE 4.2

Consider the RCBD for the vascular grafts described in Example 4.1. Suppose that we wish to determine the appropriate number of blocks to run if we are interested in detecting a true maximum difference in yield of 6 with a reasonably high probability and an estimate of the standard deviation of the errors is $\sigma = 3$. From Equation 3.45, the minimum value of Φ^2 is (writing b , the number of blocks, for n)

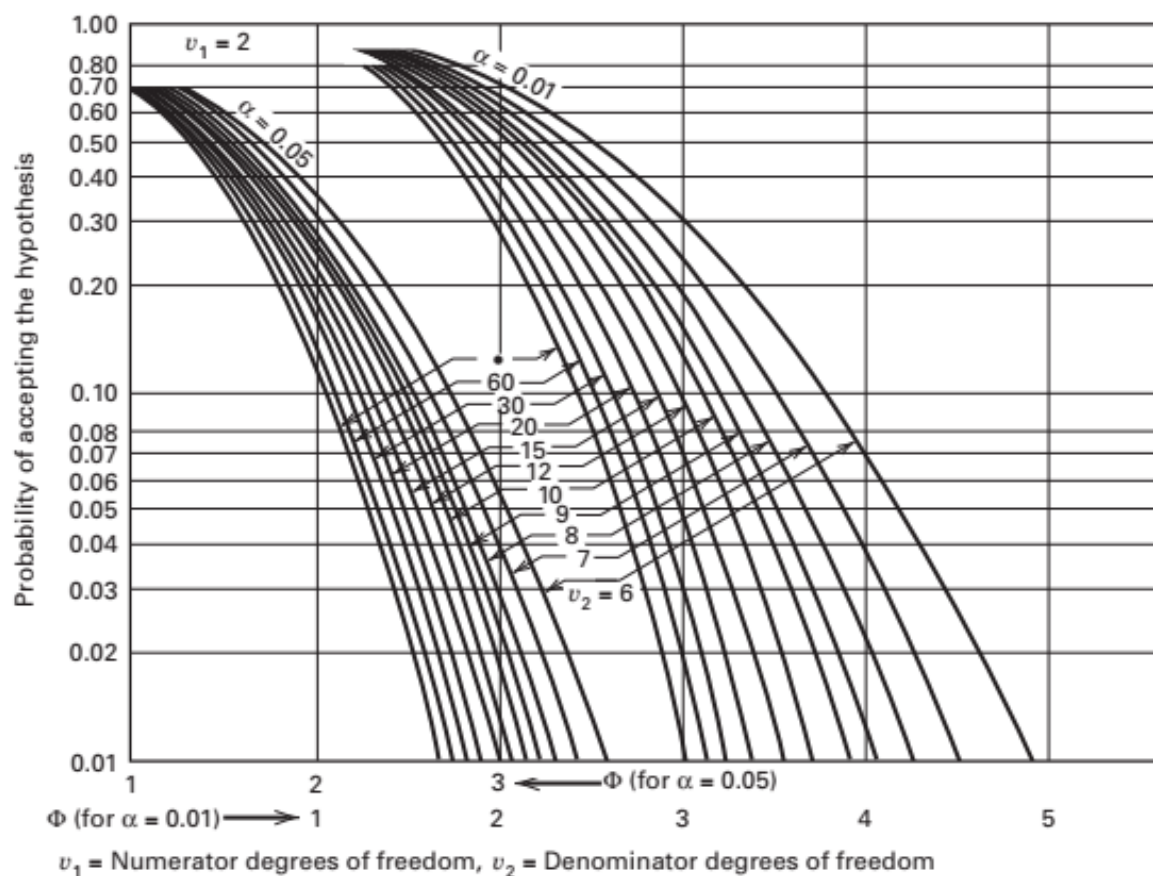
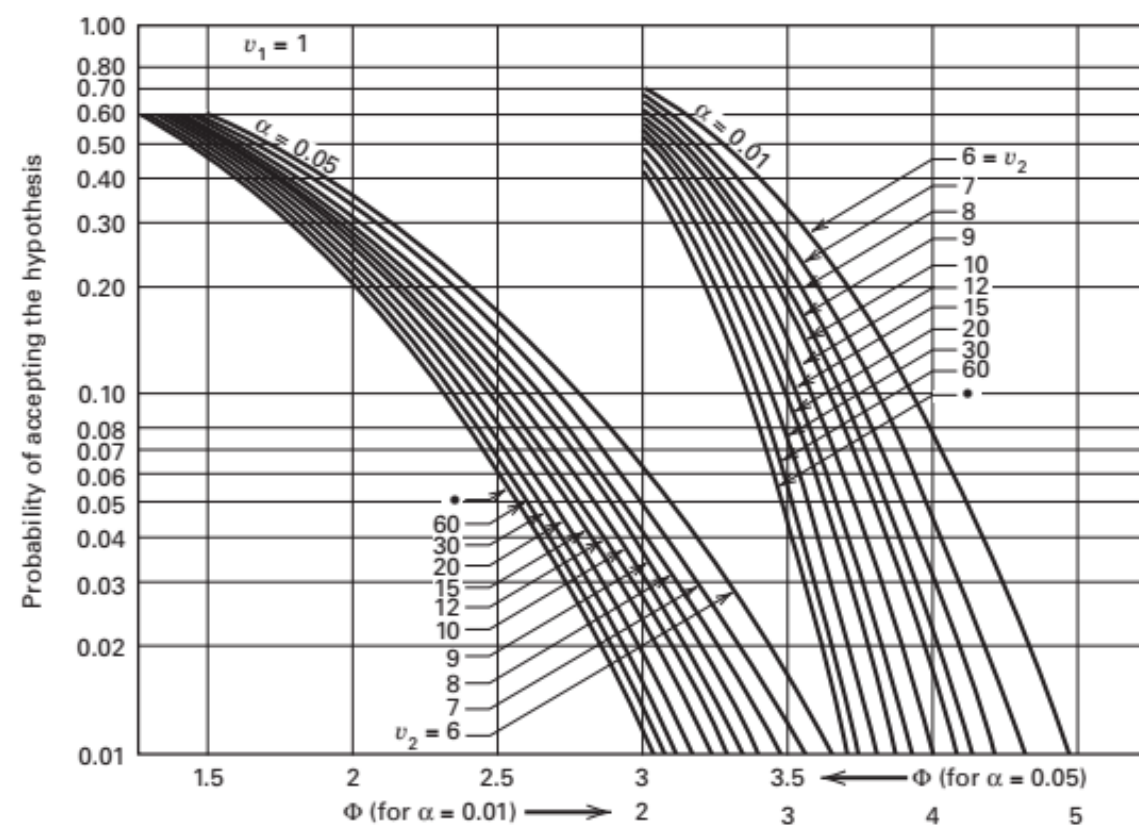
$$\Phi^2 = \frac{bD^2}{2a\sigma^2}$$

where D is the maximum difference we wish to detect. Thus,

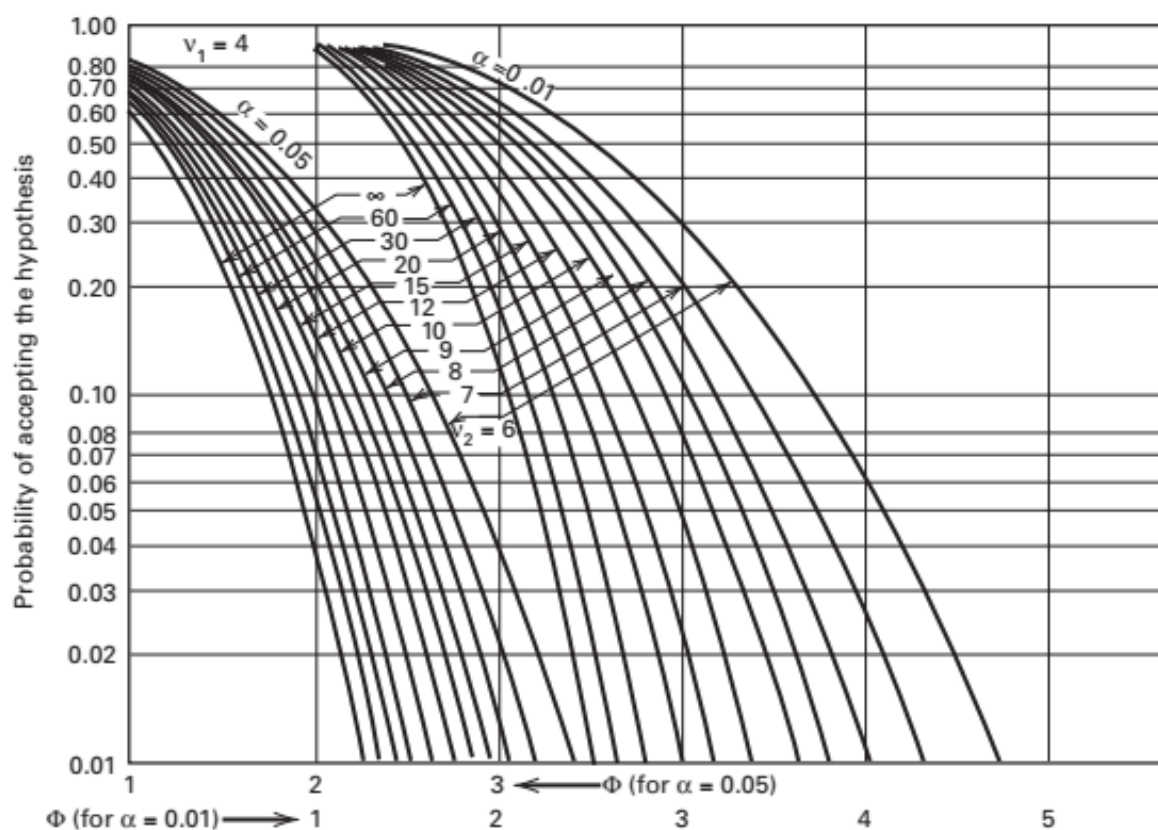
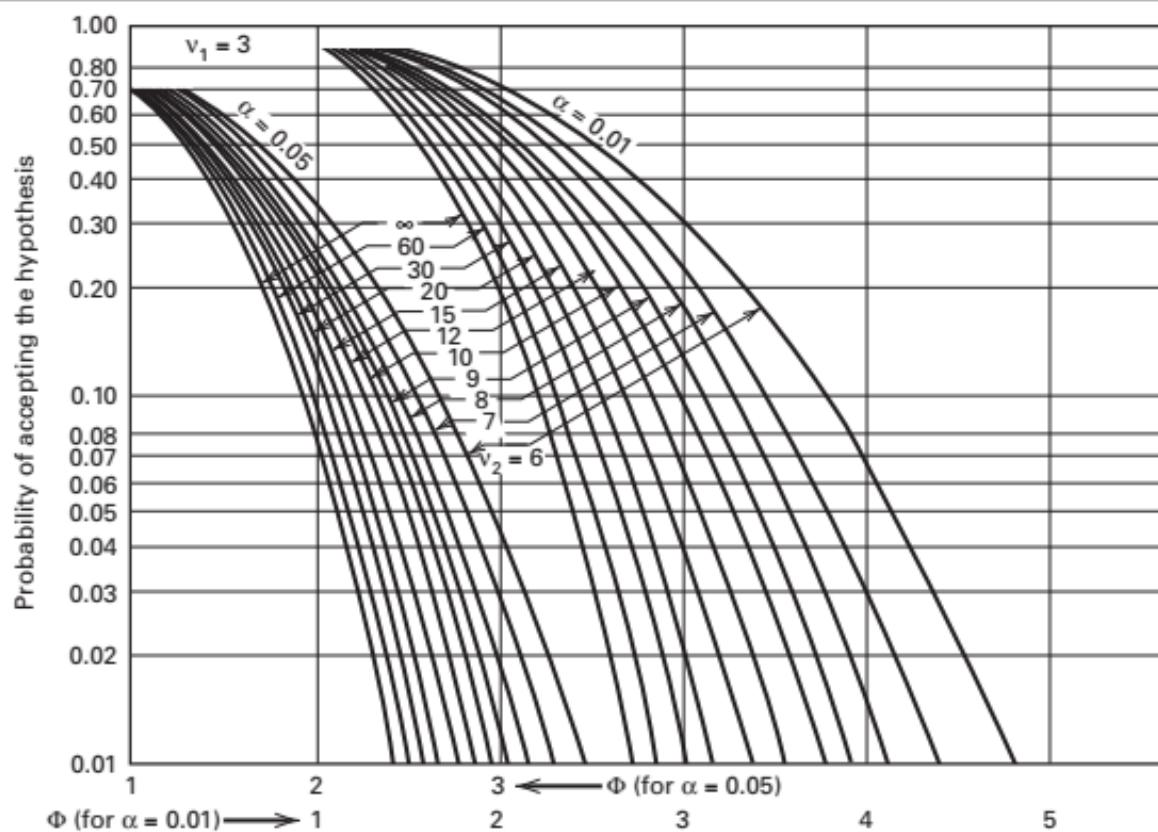
$$\Phi^2 = \frac{b(6)^2}{2(4)(3)^2} = 0.5b$$

If we use $b = 5$ blocks, $\Phi = \sqrt{0.5b} = \sqrt{0.5(5)} = 1.58$, and there are $(a - 1)(b - 1) = 3(4) = 12$ error degrees of freedom. Appendix Chart V with $\nu_1 = a - 1 = 3$ and $\alpha = 0.05$ indicates that the β risk for this design is approximately 0.55 (power = $1 - \beta = 0.45$). If we use $b = 6$ blocks, $\Phi = \sqrt{0.5b} = \sqrt{0.5(6)} = 1.73$, with $(a - 1)(b - 1) = 3(5) = 15$ error degrees of freedom, and the corresponding β risk is approximately 0.4 (power = $1 - \beta = 0.6$). Because the batches of resin are expensive and the cost of experimentation is high, the experimenter decides to use six blocks, even though the power is only about 0.6 (actually many experiments work very well with power values of only 0.5 or higher).

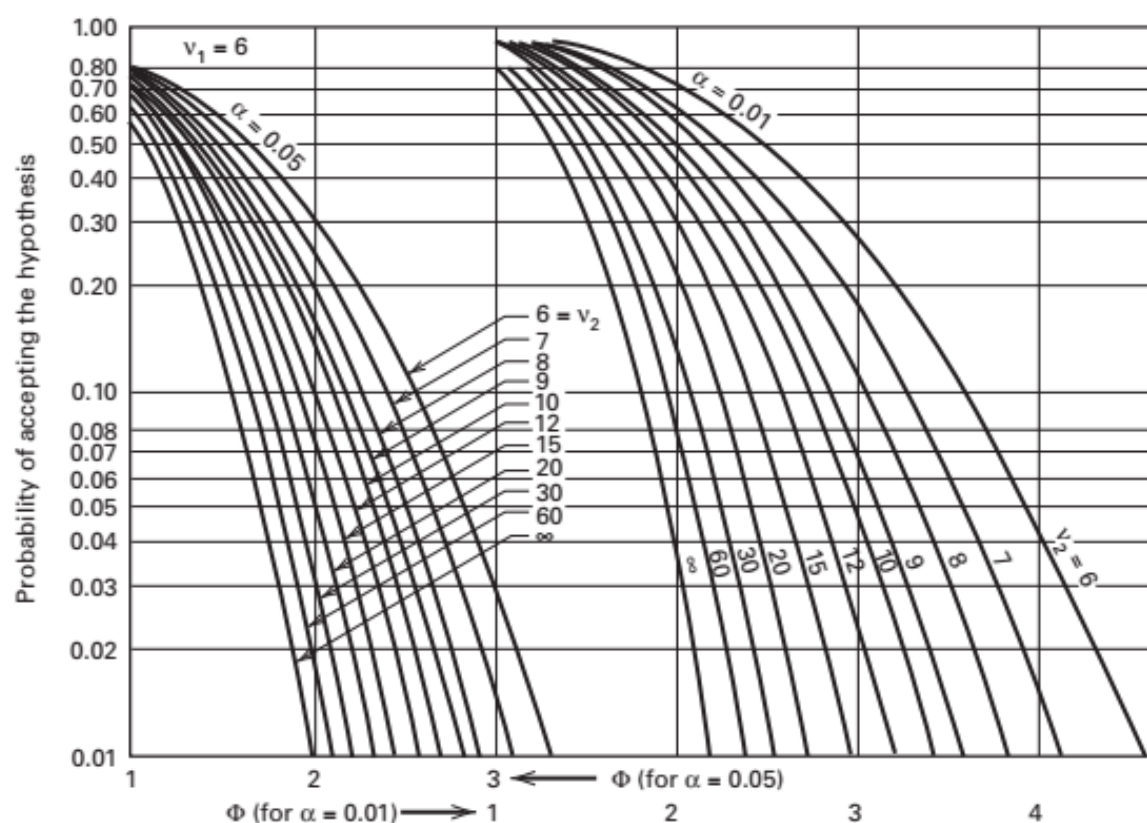
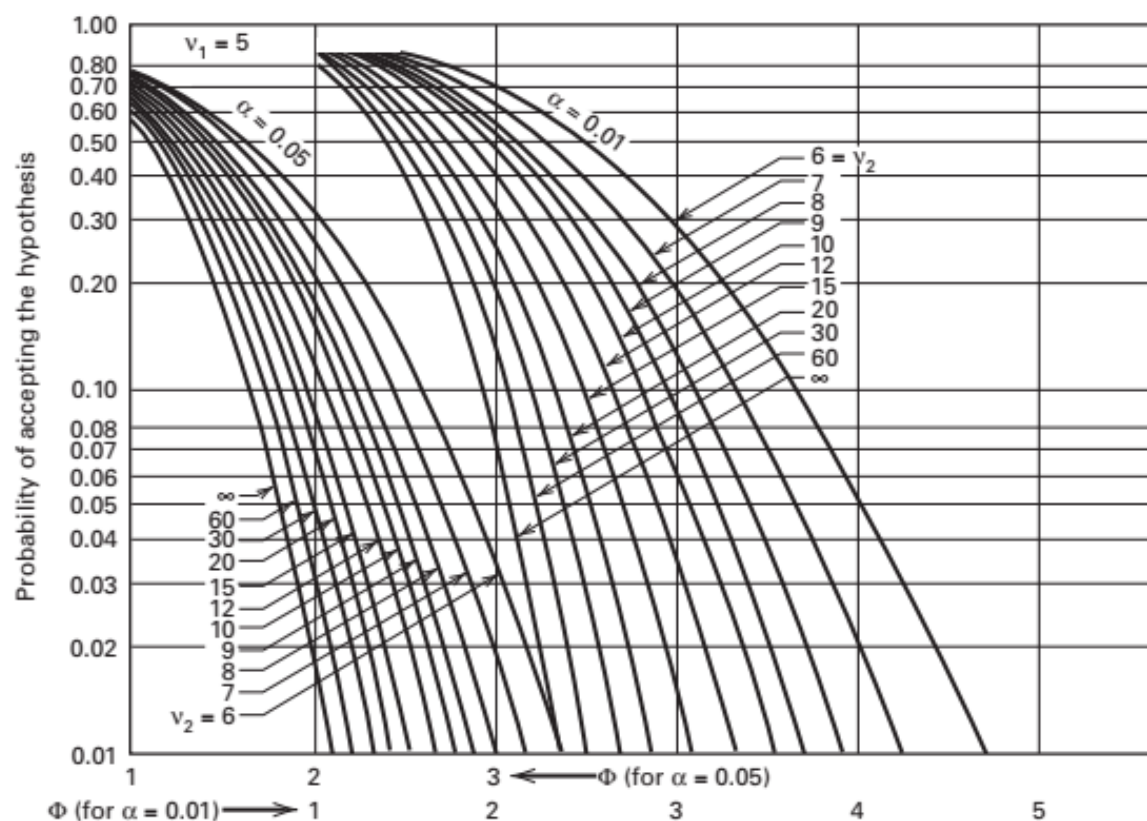
V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance^a



V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (*Continued*)



V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (*Continued*)



V Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (Continued)

